

# Agentic AI System for Enterprise Cloud Migration

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London Stock Exchange Group (LSEG)

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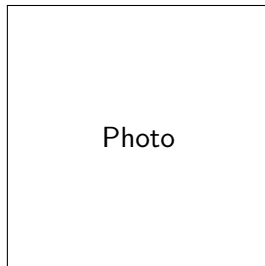


# About the Speakers



**Timothy Wong**

Tim is an AI/ML engineer at LSEG with over ten years experience in enterprise data science applications. He hosted talks at EARL 2016, 17, 19 & 24.



**Alessandro Abati**

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# LSEG

LSEG is a leading financial markets infrastructure and data provider operating connected businesses to serve customers across the entire financial markets value chain.

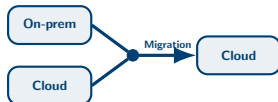
- ▶ **London Stock Exchange**
- ▶ **FTSE Russell:** Indices across asset classes, including FTSE 100, FTSE 250, Russell 2000, and ESG indices.
- ▶ **Data & Analytics:** Workspace platform, pricing, reference data.
- ▶ **Clearing:** LCH, multi-asset class clearing house.
- ▶ **Scale:** ~26 000 employees across many countries.

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# Business Requirements and Context

Co-Migrate supports the applications' journey to the cloud, turning a slow, manual migration effort into an AI-assisted one.

## Migration patterns



## From codebase to discovery report

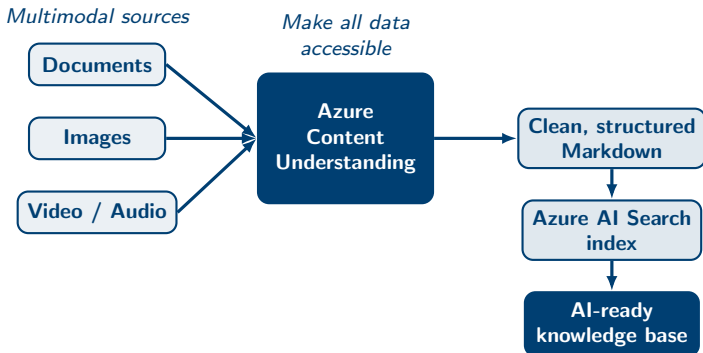


- **Discovery first:** every migration starts with a discovery stage to map dependencies and spot risks.

## The problems Co-Migrate solves

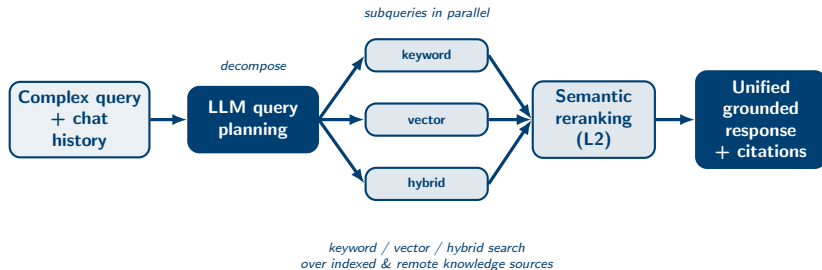
- **The bottleneck:** teams can spend up to **53 weeks** on discovery alone - slow, manual and costly.
- **Risk matters early:** spotting risks and complexity up front is key to a successful migration.

# Data Preprocessing - Making Documentation AI-Ready



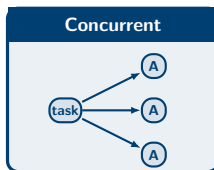
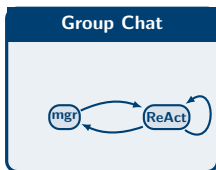
- **Content Understanding:** one analyzer for *any* modality - OCR, layout, tables and figure descriptions turn messy files into clean Markdown.
- **Make all data accessible:** charts, diagrams and scans become searchable text, so nothing is lost to downstream agents.
- **Azure AI Search:** content is chunked and indexed into an AI-ready knowledge base for agentic retrieval.

# Inside Agentic Retrieval - A Multi-Query Pipeline



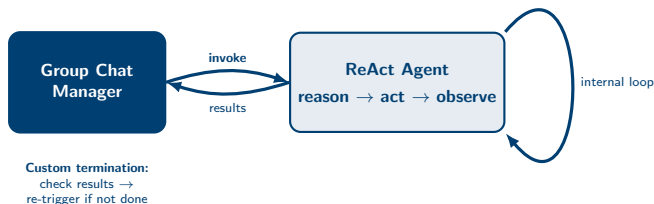
- ▶ **Beyond classic RAG:** handles multi-part questions, chat context, typos and paraphrasing that a single query can't.
- ▶ **Query planning:** an LLM breaks the query down into focused subqueries that run *in parallel*, each semantically reranked.
- ▶ **Grounded output:** best results are merged into one response with source citations and an activity log for agents.

# Agentic Orchestration - Overview



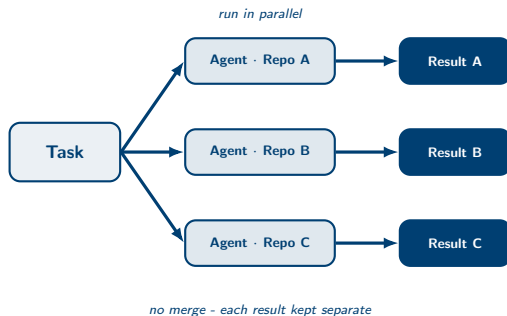
- **Framework:** Microsoft **Semantic Kernel** agent orchestration, tailored to our use case.
- **Two patterns:** *Group Chat* (1 manager + 1 ReAct agent, custom termination) and *Concurrent* (parallel, per-repo agents).
- **Writer/Reviewer** is a *Round-Robin* flavour of Group Chat - same orchestration, alternating turns.

# Agentic Orchestration - Group Chat



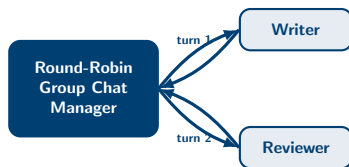
- **Our setup:** *one* manager coordinating a *single* ReAct agent - not a multi-agent debate.
- **Why a manager:** a ReAct agent's reason-act loop is trained to stop early; the manager inspects its results and *re-triggers* the loop when the task isn't complete.
- **Net effect:** a custom termination strategy on top of the agent's own reason-act loop.

# Agentic Orchestration - Concurrent



- **Our setup:** the task fans out to one agent *per repo*, all running in parallel.
- **No aggregation:** results are *not* combined - each agent only needs its own repo's context.
- **Benefits:** parallelism for speed, plus clean isolation - no cross-repo context bleed.

# Agentic Orchestration - Writer/Reviewer



## What Round-Robin overrides

`SelectNextAgent` - picks the next agent by a fixed **cyclic order** (a round-robin cursor).

*Everything else is inherited from the base `GroupChatManager`.*

- ▶ **Same pattern:** reuses the *Group Chat* orchestration with the `RoundRobinGroupChatManager` flavour - no custom manager logic needed.
- ▶ **Deterministic turns:** Writer drafts, Reviewer critiques, alternating for a fixed number of rounds (writer gets the last word).
- ▶ **Benefits:** a built-in quality gate - iterative critique lifts output quality with predictable turns and no custom manager logic.

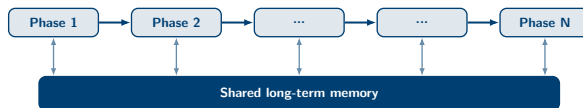
# Agent Memory

## Shared long-term memory

- ▶ A single shared memory accumulates findings across the whole run.
- ▶ Each phase reads what earlier phases discovered and writes back its own results.
- ▶ Later phases build on earlier work instead of re-reading the raw inputs.

## Passing context between phases

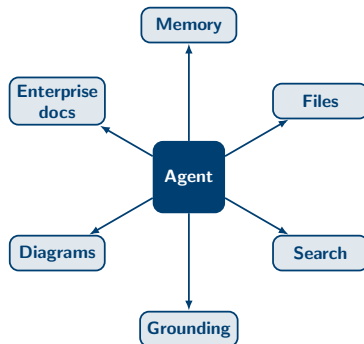
- ▶ The pipeline is serial: every phase produces a report.
- ▶ Each report, with the shared memory, becomes the input context for the next phase.
- ▶ By the final phase, agents reason over the full accumulated picture.



*Reports flow forward; every phase reads from and writes to the shared memory.*

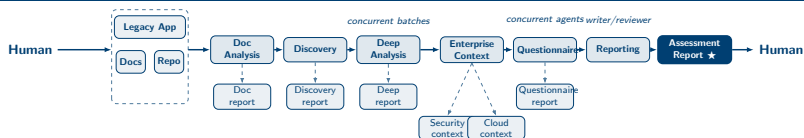
## Tools available to each agent

- ▶ **Memory:** the agent keeps long-term memory as a shared file on disk, loading past findings at start-up and writing new results back after each step so knowledge persists across the run.
- ▶ **File access:** read the codebase and documents.
- ▶ **Knowledge retrieval:** ask questions over indexed documents, with cited sources.
- ▶ **Enterprise knowledge:** search organisational security, cloud and DevOps guidance.
- ▶ **Diagrams:** create, validate and fix architecture diagrams offline.



*Each agent is given only the tools it needs.*

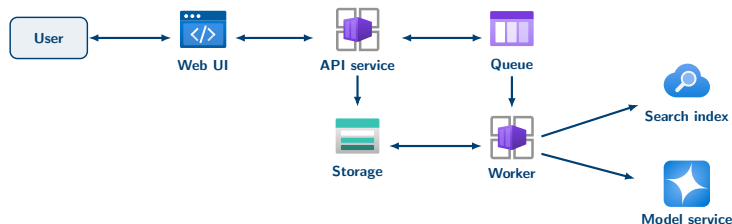
# E2E Orchestration Phases



*Inputs (documents & code) enter on the left; each phase records its findings and passes context forward.*

- ▶ **Discovery:** group the code and documents into logical components. *(Output: a map of the system.)*
- ▶ **Deep Analysis:** assess each component's migration readiness, in parallel. *(Output: per-component findings.)*
- ▶ **Enterprise Context:** apply organisational security and cloud standards. *(Output: policy-aligned recommendations.)*
- ▶ **Readiness Questionnaire:** answer a standard cloud-readiness questionnaire. *(Output: structured answers.)*
- ▶ **Reporting:** consolidate and review all findings. *(Output: the final assessment report.)*

# Infrastructure & Backend



- ▶ **Flow:** users upload through the web UI; the API stores inputs and queues a job; a worker then runs the phase pipeline.
- ▶ **Shared services:** the worker draws on storage, the search index and the model service to complete each phase.
- ▶ **Isolated environment:** the system runs fully offline, with automatic retries on transient failures.

Thank you  
Any questions?